HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed and implemented **embedded C++ firmware** for **ESP32 microcontroller** platforms with **bare-metal and FreeRTOS**-based application logic, optimizing **memory footprint by 18%** through efficient object-oriented abstraction layers and constraint-aware heap management in resource-limited environments.
- Architected **wireless communication stacks** for **Bluetooth (BLE) and Wi-Fi provisioning** with **TCP/IP security** implementation, reducing connection drop-out failures by **22%** through robust error handling, packet retry logic, and real-time monitoring on **ESP-IDF** ecosystem.
- Integrated **Modbus (RTU over RS-485 and TCP)** wired protocol drivers with external hardware sensors and industrial buses, building **register map** parsers and I2C/UART device interfaces that bridged **5+ inverter and storage system peripherals** to cloud IoT infrastructure with **zero production incidents**.
- Collaborated with hardware engineers and **cloud/backend developers** across full firmware lifecycle—from **architecture design through production deployment**—ensuring testable, maintainable device behaviour through cross-functional UAT, QA integration, and **Jenkins CI/CD pipelines** for automated firmware builds and OTA update delivery.
- Authored **Python test automation scripts** for embedded device provisioning, configuration tooling, and serial/USB interface management, reducing manual testing cycle time by **35%** and enabling **20+ daily firmware deployments** across heterogeneous **MCU families** (ESP32, STM32, Nordic nRF5x) in Linux cross-compilation environments.
- Debugged wireless protocol packets and device-level logic using **cloud IoT fundamentals** knowledge, implementing telemetry pipelines that streamed real-time sensor data and device state from **smart home and energy devices** to backend services, achieving **99.2% uptime SLA** across production fleet.

SKILLS

Embedded C/C++ – bare-metal development, FreeRTOS RTOS, memory management in constrained environments, object-oriented design patterns, clean code abstraction layers.

Microcontrollers & MCU Platforms – ESP32, ESP32-S/C/H series, ESP-IDF ecosystem, STM32, Nordic nRF5x, RP2040, flash/RAM optimization, peripheral configuration.

Wireless Protocols – Bluetooth (BLE), Wi-Fi, TCP/IP, provisioning, security, packet debugging, connection management, advertisement packet handling.

Wired Protocols & Device Integration – Modbus (RTU over RS-485 and TCP), register maps, I2C, UART, serial interfaces, hardware peripheral bridging, sensor integration.

Linux & Build Toolchains – Linux environments, shell scripting, cross-compilation, serial/USB interface management, Python automation scripting.

Cloud IoT & Production Systems – IoT infrastructure integration, telemetry pipelines, firmware deployment, OTA updates, reliability engineering, SLA management.

Development Tools & Methodologies – Git, Jenkins CI/CD, Jira, code reviews, Agile/Scrum, change-control management, testing and QA collaboration.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Smart Home Energy Device Connectivity Stack (HCL Technologies)

- Led end-to-end firmware architecture for multi-protocol **smart home energy device ecosystem**, implementing **Bluetooth Low Energy (BLE)** provisioning stack with **Wi-Fi fallback** and **TCP/IP security** across **ESP32-S3 microcontroller platform**, enabling seamless device-to-cloud enrollment for **50,000+ customer installations**.
- Designed and debugged **Modbus TCP and RTU over RS-485** integration layers for real-time communication with external inverters, battery storage systems, and solar monitoring devices, implementing robust **register map parsing**, error recovery, and timeout handling that eliminated **17 critical production incidents** related to stale device state.
- Optimized **embedded C++ application logic** and **FreeRTOS task scheduling** to achieve **35% reduction in peak power consumption** and **22% improvement in wireless throughput**, leveraging bare-metal peripheral control, efficient memory allocation, and adaptive connection retry strategies in resource-constrained environments.
- Built comprehensive **Python test automation framework** for firmware provisioning, serial interface communication, and OTA update validation across heterogeneous **ESP32, STM32, and Nordic nRF5x platforms**, reducing regression test cycles from 8 hours to **90 minutes** and enabling **daily production releases**.
- Established **cloud IoT telemetry pipeline** connecting device-level sensor data and wireless protocol state to backend analytics and customer dashboards, architecting **JSON-over-MQTT publish/subscribe messaging** with guaranteed delivery semantics, achieving **99.2% uptime SLA** and **< 500ms end-to-end latency** across all device fleet.

SWIFT Payments Operations Dashboard & Protocol Integration (HCL Technologies)

- Developed production monitoring and incident response tooling for **SWIFT messaging** and **ISO 20022 payment flows**, implementing real-time **MT/MX format validation** and **SEPA/cross-border transaction tracking** that improved payment-message visibility by **40%** and reduced mean-time-to-resolution (MTTR) for P1 incidents by **28%**.
- Collaborated with banking-domain BAs and operations teams to design **regulatory compliance** workflows incorporating **GDPR data protection**, audit trail generation, and reconciliation controls, ensuring **zero compliance violations** across **100M+ daily transactions**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)